



Chapter 5: Advanced SQL

Database System Concepts, 7th Ed.

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Outline

- Accessing SQL From a Programming Language
- Functions and Procedures
- Triggers
- Recursive Queries
- Advanced Aggregation Features



Accessing SQL from a Programming Language

A database programmer must have access to a general-purpose programming language for at least two reasons

- Not all queries can be expressed in SQL, since SQL does not provide the full expressive power of a general-purpose language.
- Non-declarative actions -- such as printing a report, interacting with a user, or sending the results of a query to a graphical user interface -- cannot be done from within SQL.



Accessing SQL from a Programming Language (Cont.)

There are two approaches to accessing SQL from a general-purpose programming language

- A general-purpose program -- can connect to and communicate with a database server using a collection of functions
- Embedded SQL -- provides a means by which a program can interact with a database server.
 - The SQL statements are translated at compile time into function calls.
 - At runtime, these function calls connect to the database using an API that provides dynamic SQL facilities.



JDBC



JDBC

- **JDBC** is a Java API for communicating with database systems supporting SQL.
- JDBC supports a variety of features for querying and updating data, and for retrieving query results.
- JDBC also supports metadata retrieval, such as querying about relations present in the database and the names and types of relation attributes.
- Model for communicating with the database:
 - Open a connection
 - Create a “statement” object
 - Execute queries using the statement object to send queries and fetch results
 - Exception mechanism to handle errors



JDBC Code

```
public static void JDBCexample(String dbid, String userid, String passwd)
{
    try (Connection conn = DriverManager.getConnection(
        "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
        Statement stmt = conn.createStatement();
    )
    {
        ... Do Actual Work ....
    }
    catch (SQLException sqle) {
        System.out.println("SQLException : " + sqle);
    }
}
```

NOTE: Above syntax works with Java 7, and JDBC 4 onwards.

Resources opened in “try (....)” syntax (“try with resources”) are automatically closed at the end of the try block



JDBC Code for Older Versions of Java/JDBC

```
public static void JDBCexample(String dbid, String userid, String passwd)
{
    try {
        Class.forName ("oracle.jdbc.driver.OracleDriver");
        Connection conn = DriverManager.getConnection(
            "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
        Statement stmt = conn.createStatement();
        ... Do Actual Work ....
        stmt.close();
        conn.close();
    }
    catch (SQLException sqle) {
        System.out.println("SQLException : " + sqle);
    }
}
```

NOTE: Class.forName is not required from JDBC 4 onwards. The try with resources syntax in prev slide is preferred for Java 7 onwards.



JDBC Code (Cont.)

- Update to database

```
try {  
    stmt.executeUpdate(  
        "insert into instructor values('77987', 'Kim', 'Physics', 98000)");  
} catch (SQLException sqle)  
{  
    System.out.println("Could not insert tuple. " + sqle);  
}
```

- Execute query and fetch and print results

```
ResultSet rset = stmt.executeQuery(  
    "select dept_name, avg (salary)  
    from instructor  
    group by dept_name");  
while (rset.next()) {  
    System.out.println(rset.getString("dept_name") + " " +  
        rset.getFloat(2));  
}
```



JDBC SUBSECTIONS

- Connecting to the Database
- Shipping SQL Statements to the Database System
- Exceptions and Resource Management
- Retrieving the Result of a Query
- Prepared Statements
- Callable Statements
- Metadata Features
- Other Features
- Database Access from Python



JDBC Code Details

- Getting result fields:
 - **`rs.getString("dept_name")` and `rs.getString(1)` equivalent if `dept_name` is the first argument of select result.**
- Dealing with Null values
**`int a = rs.getInt("a");`
`if (rs.isNull()) Systems.out.println("Got null value");`**



Prepared Statement

- `PreparedStatement pstmt = conn.prepareStatement(
"insert into instructor values(?,?,?,?)");

pstmt.setString(1, "88877");
pstmt.setString(2, "Perry");
pstmt.setString(3, "Finance");
pstmt.setInt(4, 125000);
pstmt.executeUpdate();
pstmt.setString(1, "88878");
pstmt.executeUpdate();`
- WARNING: always use prepared statements when taking an input from the user and adding it to a query
 - NEVER create a query by concatenating strings
 - `"insert into instructor values(' " + ID + " ', ' " + name + " ', " + "' + dept name + " ', "' + balance + ')"`
 - What if name is "D'Souza"?



SQL Injection

- Suppose query is constructed using
 - "select * from instructor where name = '" + name + "'"
- Suppose the user, instead of entering a name, enters:
 - X' or 'Y' = 'Y
- then the resulting statement becomes:
 - "select * from instructor where name = '" + "X' or 'Y' = 'Y" + "'"
 - which is:
 - select * from instructor where name = 'X' or 'Y' = 'Y'
 - User could have even used
 - X'; update instructor set salary = salary + 10000; --
- Prepared statement internally uses:
"select * from instructor where name = 'X\' or \'Y\' = \'Y\'"
 - **Always use prepared statements, with user inputs as parameters**



Metadata Features

- ResultSet metadata
- E.g. after executing query to get a ResultSet rs:
 - ```
ResultSetMetaData rsmd = rs.getMetaData();
for(int i = 1; i <= rsmd.getColumnCount(); i++) {
 System.out.println(rsmd.getColumnName(i));
 System.out.println(rsmd.getColumnTypeName(i));
}
```
- How is this useful?



# Metadata (Cont)

- Database metadata
- DatabaseMetaData dbmd = conn.getMetaData();  
// Arguments to getColumnns: Catalog, Schema-pattern, Table-pattern,  
// and Column-Pattern  
// Returns: One row for each column; row has a number of attributes  
// such as COLUMN\_NAME, TYPE\_NAME  
// The value null indicates all Catalogs/Schemas.  
// The value "" indicates current catalog/schema  
// The value "%" has the same meaning as SQL **like** clause  
ResultSet rs = dbmd.getColumnns(null, "univdb", "department", "%");  
while( rs.next()) {  
    System.out.println(rs.getString("COLUMN\_NAME"),  
                        rs.getString("TYPE\_NAME");  
}  
■ And where is this useful?



# Metadata (Cont)

- Database metadata
- DatabaseMetaData dbmd = conn.getMetaData();  
// Arguments to getTables: Catalog, Schema-pattern, Table-pattern,  
// and Table-Type  
// Returns: One row for each table; row has a number of attributes  
// such as TABLE\_NAME, TABLE\_CAT, TABLE\_TYPE, ..  
// The value null indicates all Catalogs/Schemas.  
// The value "" indicates current catalog/schema  
// The value "%" has the same meaning as SQL **like** clause  
// The last attribute is an array of types of tables to return.  
// TABLE means only regular tables  
ResultSet rs = dbmd.getTables ("", "", "%", new String[] {"TABLES"});  
while( rs.next()) {  
    System.out.println(rs.getString("TABLE\_NAME"));  
}  
■ And where is this useful?





# Finding Primary Keys

- DatabaseMetaData dmd = connection.getMetaData();

// Arguments below are: Catalog, Schema, and Table

// The value "" for Catalog/Schema indicates current catalog/schema

// The value null indicates all catalogs/schemas

ResultSet rs = dmd.getPrimaryKeys("", "", tableName);

while(rs.next()){

    // KEY\_SEQ indicates the position of the attribute in

    // the primary key, which is required if a primary key has multiple

    // attributes

    System.out.println(rs.getString("KEY\_SEQ"),  
                            rs.getString("COLUMN\_NAME");

}



# Transaction Control in JDBC

- By default, each SQL statement is treated as a separate transaction that is committed automatically
  - bad idea for transactions with multiple updates
- Can turn off automatic commit on a connection
  - `conn.setAutoCommit(false);`
- Transactions must then be committed or rolled back explicitly
  - `conn.commit();`    or
  - `conn.rollback();`
- `conn.setAutoCommit(true)` turns on automatic commit.



# Other JDBC Features

- Calling functions and procedures
  - `CallableStatement cStmt1 = conn.prepareCall("{? = call some function(?)})");`
  - `CallableStatement cStmt2 = conn.prepareCall("{call some procedure(?,?)})");`
- Handling large object types
  - `getBlob()` and `getClob()` that are similar to the `getString()` method, but return objects of type `Blob` and `Clob`, respectively
  - get data from these objects by `getBytes()`
  - associate an open stream with Java `Blob` or `Clob` object to update large objects
    - `blob.setBlob(int parameterIndex, InputStream inputStream).`



# JDBC Resources

- JDBC Basics Tutorial
  - <https://docs.oracle.com/javase/tutorial/jdbc/index.html>



# SQLJ

- JDBC is overly dynamic, errors cannot be caught by compiler
- SQLJ: embedded SQL in Java
  - ```
#sql iterator deptInfoIter ( String dept name, int avgSal);  
deptInfoIter iter = null;  
#sql iter = { select dept_name, avg(salary) from instructor  
              group by dept name };  
while (iter.next()) {  
    String deptName = iter.dept_name();  
    int avgSal = iter.avgSal();  
    System.out.println(deptName + " " + avgSal);  
}  
iter.close();
```



ODBC



ODBC

- Open DataBase Connectivity (ODBC) standard
 - standard for application program to communicate with a database server.
 - application program interface (API) to
 - open a connection with a database,
 - send queries and updates,
 - get back results.
- Applications such as GUI, spreadsheets, etc. can use ODBC



Embedded SQL

- The SQL standard defines embeddings of SQL in a variety of programming languages such as C, C++, Java, Fortran, and PL/1,
- A language to which SQL queries are embedded is referred to as a **host language**, and the SQL structures permitted in the host language comprise *embedded SQL*.
- The basic form of these languages follows that of the System R embedding of SQL into PL/1.
- **EXEC SQL** statement is used in the host language to identify embedded SQL request to the preprocessor

EXEC SQL <embedded SQL statement >;

Note: this varies by language:

- In some languages, like COBOL, the semicolon is replaced with END-EXEC
- In Java embedding uses `# SQL { .... };`



Embedded SQL (Cont.)

- Before executing any SQL statements, the program must first connect to the database. This is done using:

EXEC-SQL **connect to** *server* **user** *user-name* **using** *password*;

Here, *server* identifies the server to which a connection is to be established.

- Variables of the host language can be used within embedded SQL statements. They are preceded by a colon (:) to distinguish from SQL variables (e.g., *:credit_amount*)
- Variables used as above must be declared within DECLARE section, as illustrated below. The syntax for declaring the variables, however, follows the usual host language syntax.

EXEC-SQL BEGIN DECLARE SECTION}

 int *credit-amount* ;

EXEC-SQL END DECLARE SECTION;



Embedded SQL (Cont.)

- To write an embedded SQL query, we use the
declare *c* cursor for <SQL query>
statement. The variable *c* is used to identify the query
- Example:
 - From within a host language, find the ID and name of students who have completed more than the number of credits stored in variable *credit_amount* in the host language
 - Specify the query in SQL as follows:

EXEC SQL

```
declare c cursor for  
select ID, name  
from student  
where tot_cred > :credit_amount
```

END_EXEC



Embedded SQL (Cont.)

- The **open** statement for our example is as follows:

```
EXEC SQL open c ;
```

This statement causes the database system to execute the query and to save the results within a temporary relation. The query uses the value of the host-language variable *credit-amount* at the time the **open** statement is executed.

- The fetch statement causes the values of one tuple in the query result to be placed on host language variables.

```
EXEC SQL fetch c into :si, :sn END_EXEC
```

Repeated calls to fetch get successive tuples in the query result



Embedded SQL (Cont.)

- A variable called SQLSTATE in the SQL communication area (SQLCA) gets set to '02000' to indicate no more data is available
- The **close** statement causes the database system to delete the temporary relation that holds the result of the query.

EXEC SQL close *c* ;

Note: above details vary with language. For example, the Java embedding defines Java iterators to step through result tuples.



Updates Through Embedded SQL

- Embedded SQL expressions for database modification (**update**, **insert**, and **delete**)
- Can update tuples fetched by cursor by declaring that the cursor is for update

EXEC SQL

```
declare c cursor for  
  select *  
  from instructor  
  where dept_name = 'Music'  
  for update
```

- We then iterate through the tuples by performing **fetch** operations on the cursor (as illustrated earlier), and after fetching each tuple we execute the following code:

```
update instructor  
  set salary = salary + 1000  
  where current of c
```



Functions and Procedures



Functions and Procedures

- Functions and procedures allow “business logic” to be stored in the database and executed from SQL statements.
- These can be defined either by the procedural component of SQL or by an external programming language such as Java, C, or C++.
- The syntax we present here is defined by the SQL standard.
 - Most databases implement nonstandard versions of this syntax.



Declaring SQL Functions

- Define a function that, given the name of a department, returns the count of the number of instructors in that department.

```
create function dept_count (dept_name varchar(20))  
  returns integer  
  begin  
    declare d_count integer;  
    select count ( * ) into d_count  
    from instructor  
    where instructor.dept_name = dept_name  
    return d_count;  
  end
```

- The function *dept_count* can be used to find the department names and budget of all departments with more than 12 instructors.

```
select dept_name, budget  
from department  
where dept_count (dept_name ) > 12
```




Table Functions

- The SQL standard supports functions that can return tables as results; such functions are called **table functions**
- Example: Return all instructors in a given department

create function *instructor_of*(*dept_name* **char**(20))

returns table (

ID **varchar**(5),
name **varchar**(20),
dept_name **varchar**(20),
salary **numeric**(8,2))

return table

(**select** *ID*, *name*, *dept_name*, *salary*
from *instructor*
where *instructor.dept_name* = *instructor_of.dept_name*)

- Usage

select *
from table (*instructor_of*('Music'))



SQL Procedures

- The *dept_count* function could instead be written as procedure:

```
create procedure dept_count_proc (in dept_name varchar(20),  
                                out d_count integer)
```

```
begin
```

```
    select count(*) into d_count
```

```
    from instructor
```

```
    where instructor.dept_name = dept_count_proc.dept_name
```

```
end
```

- The keywords **in** and **out** are parameters that are expected to have values assigned to them and parameters whose values are set in the procedure in order to return results.
- Procedures can be invoked either from an SQL procedure or from embedded SQL, using the **call** statement.

```
declare d_count integer;
```

```
call dept_count_proc( 'Physics', d_count);
```



SQL Procedures (Cont.)

- Procedures and functions can be invoked also from dynamic SQL
- SQL allows more than one procedure of the so long as the number of arguments of the procedures with the same name is different.
- The name, along with the number of arguments, is used to identify the procedure.



Language Constructs for Procedures & Functions

- SQL supports constructs that gives it almost all the power of a general-purpose programming language.
 - Warning: most database systems implement their own variant of the standard syntax below.
- Compound statement: **begin ... end**,
 - May contain multiple SQL statements between **begin** and **end**.
 - Local variables can be declared within a compound statements
- While and repeat statements:
 - **while** boolean expression **do**
 sequence of statements ;
end while
 - **repeat**
 sequence of statements ;
 until boolean expression
end repeat



Language Constructs (Cont.)

- **For** loop
 - Permits iteration over all results of a query
- Example: Find the budget of all departments

```
declare n integer default 0;  
for r as  
    select budget from department where  
dept_name = 'Music'  
do  
    set n = n + r.budget  
end for
```



Language Constructs – if-then-else

- Conditional statements (**if-then-else**)
 - if** *boolean expression*
 - then** *statement or compound statement*
 - elseif** *boolean expression*
 - then** *statement or compound statement*
 - else** *statement or compound statement*
 - end if**



Example procedure

- Registers student after ensuring classroom capacity is not exceeded
 - Returns 0 on success and -1 if capacity is exceeded
 - See book (page 202) for details
- Signaling of exception conditions, and declaring handlers for exceptions

```
declare out_of_classroom_seats condition
declare exit handler for out_of_classroom_seats
begin
...
end
```
- The statements between the **begin** and the **end** can raise an exception by executing “**signal** *out_of_classroom_seats*”
- The handler says that if the condition arises the action to be taken is to exit the enclosing the **begin end** statement.



External Language Routines

- SQL allows us to define functions in a programming language such as Java, C#, C or C++.
 - Can be more efficient than functions defined in SQL, and computations that cannot be carried out in SQL can be executed by these functions.
- Declaring external language procedures and functions

```
create procedure dept_count_proc(in dept_name varchar(20),  
                                out count integer)
```

```
language C
```

```
external name '/usr/avi/bin/dept_count_proc'
```

```
create function dept_count(dept_name varchar(20))
```

```
returns integer
```

```
language C
```

```
external name '/usr/avi/bin/dept_count'
```




External Language Routines (Cont.)

- Benefits of external language functions/procedures:
 - more efficient for many operations, and more expressive power.
- Drawbacks
 - Code to implement function may need to be loaded into database system and executed in the database system's address space.
 - risk of accidental corruption of database structures
 - security risk, allowing users access to unauthorized data
 - There are alternatives, which give good security at the cost of potentially worse performance.
 - Direct execution in the database system's space is used when efficiency is more important than security.



Security with External Language Routines

- To deal with security problems, we can do one of the following:
 - Use **sandbox** techniques
 - That is, use a safe language like Java, which cannot be used to access/damage other parts of the database code.
 - Run external language functions/procedures in a separate process, with no access to the database process' memory.
 - Parameters and results communicated via inter-process communication
- Both have performance overheads
- Many database systems support both above approaches as well as direct executing in database system address space.



Triggers



Triggers

- A **trigger** is a statement that is executed automatically by the system as a side effect of a modification to the database.
- To design a trigger mechanism, we must:
 - Specify the conditions under which the trigger is to be executed.
 - Specify the actions to be taken when the trigger executes.
- Triggers introduced to SQL standard in SQL:1999, but supported even earlier using non-standard syntax by most databases.
 - Syntax illustrated here may not work exactly on your database system; check the system manuals



Triggering Events and Actions in SQL

- Triggering event can be **insert**, **delete** or **update**
- Triggers on update can be restricted to specific attributes
 - For example, **after update of *takes* on *grade***
- Values of attributes before and after an update can be referenced
 - **referencing old row as** : for deletes and updates
 - **referencing new row as** : for inserts and updates
- Triggers can be activated before an event, which can serve as extra constraints.
For example, convert blank grades to null.

```
create trigger setnull_trigger before update of takes  
referencing new row as nrow  
for each row  
  when (nrow.grade = ' ')  
    begin atomic  
      set nrow.grade = null;  
    end;
```



Trigger to Maintain `credits_earned` value

- **create trigger *credits_earned* after update of *takes* on (*grade*)**
referencing new row as *nrow*
referencing old row as *orow*
for each row
when *nrow.grade* $\neq$ 'F' and *nrow.grade* is not null
and (*orow.grade* = 'F' or *orow.grade* is null)
begin atomic
update *student*
set *tot_cred* = *tot_cred* +
(select *credits*
from *course*
where *course.course_id* = *nrow.course_id*)
where *student.id* = *nrow.id*;
end;



Statement Level Triggers

- Instead of executing a separate action for each affected row, a single action can be executed for all rows affected by a transaction
 - Use **for each statement** instead of **for each row**
 - Use **referencing old table** or **referencing new table** to refer to temporary tables (called *transition tables*) containing the affected rows
 - Can be more efficient when dealing with SQL statements that update a large number of rows



When Not To Use Triggers

- Triggers were used earlier for tasks such as
 - Maintaining summary data (e.g., total salary of each department)
 - Replicating databases by recording changes to special relations (called **change** or **delta** relations) and having a separate process that applies the changes over to a replica
- There are better ways of doing these now:
 - Databases today provide built in materialized view facilities to maintain summary data
 - Databases provide built-in support for replication
- Risk of unintended execution of triggers, for example, when
 - Loading data from a backup copy
 - Replicating updates at a remote site
 - Trigger execution can be disabled before such actions.
- Other risks with triggers:
 - Error leading to failure of critical transactions that set off the trigger
 - Cascading execution



Home Exercise—

Each hospital can book 200 appointments for vaccination in a day. Check whether a new appointment is available in a hospital on a given date.



Recursive Queries



Recursion in SQL

- SQL:1999 permits recursive view definition
- Example: find which courses are a prerequisite, whether directly or indirectly, for a specific course

```
with recursive rec_prereq(course_id, prereq_id) as (  
    select course_id, prereq_id  
    from prereq  
    union  
    select rec_prereq.course_id, prereq.prereq_id,  
    from rec_rereq, prereq  
    where rec_prereq.prereq_id = prereq.course_id  
    )  
select *  
from rec_prereq;
```

This example view, *rec_prereq*, is called the *transitive closure* of the *prereq* relation



The Power of Recursion

- Recursive views make it possible to write queries, such as transitive closure queries, that cannot be written without recursion or iteration.
 - Intuition: Without recursion, a non-recursive non-iterative program can perform only a fixed number of joins of *prereq* with itself
 - This can give only a fixed number of levels of managers
 - Given a fixed non-recursive query, we can construct a database with a greater number of levels of prerequisites on which the query will not work
 - Alternative: write a procedure to iterate as many times as required
 - See procedure *findAllPrereqs* in book



The Power of Recursion

- Computing transitive closure using iteration, adding successive tuples to *rec_prereq*
 - The next slide shows a *prereq* relation
 - Each step of the iterative process constructs an extended version of *rec_prereq* from its recursive definition.
 - The final result is called the *fixed point* of the recursive view definition.
- Recursive views are required to be **monotonic**. That is, if we add tuples to *prereq* the view *rec_prereq* contains all of the tuples it contained before, plus possibly more



Example of Fixed-Point Computation

<i>course_id</i>	<i>prereq_id</i>
BIO-301	BIO-101
BIO-399	BIO-101
CS-190	CS-101
CS-315	CS-190
CS-319	CS-101
CS-319	CS-315
CS-347	CS-319

<i>Iteration Number</i>	<i>Tuples in c1</i>
0	
1	(CS-319)
2	(CS-319), (CS-315), (CS-101)
3	(CS-319), (CS-315), (CS-101), (CS-190)
4	(CS-319), (CS-315), (CS-101), (CS-190)
5	done



Advanced Aggregation Features



Ranking

- Ranking is done in conjunction with an order by specification.
- Suppose we are given a relation
student_grades(*ID*, *GPA*)
giving the grade-point average of each student
- Find the rank of each student.
- ```
select ID, rank() over (order by GPA desc) as s_rank
from student_grades
```
- An extra **order by** clause is needed to get them in sorted order  

```
select ID, rank() over (order by GPA desc) as s_rank
from student_grades
order by s_rank
```
- Ranking may leave gaps: e.g. if 2 students have the same top GPA, both have rank 1, and the next rank is 3
  - dense\_rank** does not leave gaps, so next dense rank would be 2





# Ranking

- Ranking can be done using basic SQL aggregation, but resultant query is very inefficient

```
select ID, (1 + (select count(*)
 from student_grades B
 where B.GPA > A.GPA)) as s_rank
from student_grades A
order by s_rank;
```



## Ranking (Cont.)

- Ranking can be done within partition of the data.
- “Find the rank of students within each department.”

```
select ID, dept_name,
 rank () over (partition by dept_name order by GPA desc)
 as dept_rank
from dept_grades
order by dept_name, dept_rank;
```

- Multiple **rank** clauses can occur in a single **select** clause.
- Ranking is done *after* applying **group by** clause/aggregation
- Can be used to find top-n results
  - More general than the **limit** *n* clause supported by many databases, since it allows top-n within each partition



# Ranking (Cont.)

- Other ranking functions:
  - **percent\_rank** (within partition, if partitioning is done)
  - **cume\_dist** (cumulative distribution)
    - fraction of tuples with preceding values
  - **row\_number** (non-deterministic in presence of duplicates)
- SQL:1999 permits the user to specify **nulls first** or **nulls last**

**select** *ID*,

**rank ( ) over (order by** *GPA* **desc nulls last)** **as** *s\_rank*  
**from** *student\_grades*



## Ranking (Cont.)

- For a given constant  $n$ , the ranking the function  $ntile(n)$  takes the tuples in each partition in the specified order, and divides them into  $n$  buckets with equal numbers of tuples.
- E.g.,  
***select ID, ntile(4) over (order by GPA desc) as quartile***  
***from student\_grades;***



# Windowing

- Used to smooth out random variations.
- E.g., **moving average**: “Given sales values for each date, calculate for each date the average of the sales on that day, the previous day, and the next day”
- **Window specification** in SQL:
  - Given relation *sales*(*date*, *value*)  
**select** *date*, **sum**(*value*) **over**  
    (**order by** *date* **between** rows 1 **preceding** and 1 **following**)  
**from** *sales*



# Windowing

- Examples of other window specifications:
  - **between rows unbounded preceding and current**
  - **rows unbounded preceding**
  - **range between 10 preceding and current row**
    - All rows with values between current row value  $-10$  to current value
  - **range interval 10 day preceding**
    - Not including current row



# Windowing (Cont.)

- Can do windowing within partitions
- E.g., Given a relation *transaction* (*account\_number*, *date\_time*, *value*), where *value* is positive for a deposit and negative for a withdrawal
  - “Find total balance of each account after each transaction on the account”

```
select account_number, date_time,
 sum (value) over
 (partition by account_number
 order by date_time
 rows unbounded preceding)
 as balance
from transaction
order by account_number, date_time
```



# OLAP





# Data Analysis and OLAP

- **Online Analytical Processing (OLAP)**
  - Interactive analysis of data, allowing data to be summarized and viewed in different ways in an online fashion (with negligible delay)
- Data that can be modeled as dimension attributes and measure attributes are called **multidimensional data**.
  - **Measure attributes**
    - measure some value
    - can be aggregated upon
    - e.g., the attribute *number* of the *sales* relation
  - **Dimension attributes**
    - define the dimensions on which measure attributes (or aggregates thereof) are viewed
    - e.g., attributes *item\_name*, *color*, and *size* of the *sales* relation



# Example sales relation

| <i>item_name</i> | <i>color</i> | <i>clothes_size</i> | <i>quantity</i> |
|------------------|--------------|---------------------|-----------------|
| skirt            | dark         | small               | 2               |
| skirt            | dark         | medium              | 5               |
| skirt            | dark         | large               | 1               |
| skirt            | pastel       | small               | 11              |
| skirt            | pastel       | medium              | 9               |
| skirt            | pastel       | large               | 15              |
| skirt            | white        | small               | 2               |
| skirt            | white        | medium              | 5               |
| skirt            | white        | large               | 3               |
| dress            | dark         | small               | 2               |
| dress            | dark         | medium              | 6               |
| dress            | dark         | large               | 12              |
| dress            | pastel       | small               | 4               |
| dress            | pastel       | medium              | 3               |
| dress            | pastel       | large               | 3               |
| dress            | white        | small               | 2               |
| dress            | white        | medium              | 3               |
| dress            | white        | large               | 0               |
| shirt            | dark         | small               | 2               |
| shirt            | dark         | medium              | 4               |

... ... ...

... ... 5.66 ...



# Cross Tabulation of *sales* by *item\_name* and *color*

*clothes\_size* **all**

|                  |       | <i>color</i> |        |       |       |
|------------------|-------|--------------|--------|-------|-------|
|                  |       | dark         | pastel | white | total |
| <i>item_name</i> | skirt | 8            | 35     | 10    | 53    |
|                  | dress | 20           | 10     | 5     | 35    |
|                  | shirt | 14           | 7      | 28    | 49    |
|                  | pants | 20           | 2      | 5     | 27    |
|                  | total | 62           | 54     | 48    | 164   |

- The table above is an example of a **cross-tabulation** (**cross-tab**), also referred to as a **pivot-table**.
  - Values for one of the dimension attributes form the row headers
  - Values for another dimension attribute form the column headers
  - Other dimension attributes are listed on top
  - Values in individual cells are (aggregates of) the values of the dimension attributes that specify the cell.



# Data Cube

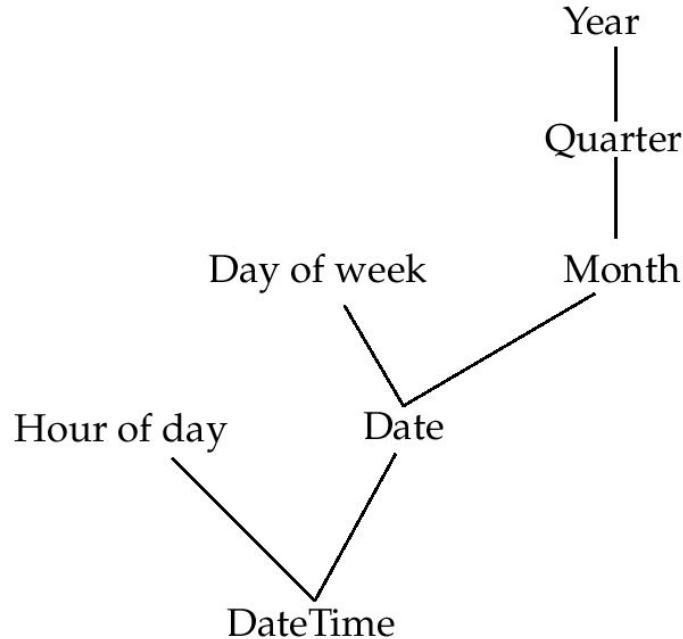
- A **data cube** is a multidimensional generalization of a cross-tab
- Can have  $n$  dimensions; we show 3 below
- Cross-tabs can be used as views on a data cube

|       |        | item_name |       |       |       |     |   |
|-------|--------|-----------|-------|-------|-------|-----|---|
|       |        | skirt     | dress | shirt | pants | all |   |
| color | dark   | 8         | 20    | 14    | 20    | 62  | 3 |
|       | pastel | 35        | 10    | 7     | 2     | 54  | 2 |
|       | white  | 10        | 8     | 28    | 5     | 48  | 7 |
|       | all    | 53        | 38    | 49    | 27    | 164 | a |

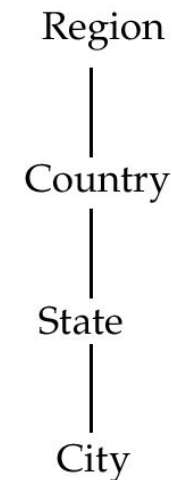


# Hierarchies on Dimensions

- **Hierarchy** on dimension attributes: lets dimensions to be viewed at different levels of detail
  - E.g., the dimension DateTime can be used to aggregate by hour of day, date, day of week, month, quarter or year



a) Time Hierarchy



b) Location Hierarchy



# Cross Tabulation With Hierarchy

- Cross-tabs can be easily extended to deal with hierarchies
  - Can drill down or roll up on a hierarchy

*clothes\_size:* **all**

|            |          | <i>category</i> <i>item_name</i> <i>color</i> |        |       |       |     |
|------------|----------|-----------------------------------------------|--------|-------|-------|-----|
|            |          | dark                                          | pastel | white | total |     |
| womenswear | skirt    | 8                                             | 8      | 10    | 53    | 88  |
|            | dress    | 20                                            | 20     | 5     | 35    |     |
|            | subtotal | 28                                            | 28     | 15    |       |     |
| menswear   | pants    | 14                                            | 14     | 28    | 49    | 76  |
|            | shirt    | 20                                            | 20     | 5     | 27    |     |
|            | subtotal | 34                                            | 34     | 33    |       |     |
| total      |          | 62                                            | 62     | 48    |       | 164 |



# Relational Representation of Cross-tabs

- Cross-tabs can be represented as relations
  - We use the value **all** is used to represent aggregates.
  - The SQL standard actually uses null values in place of **all** despite confusion with regular null values.

| <i>item_name</i> | <i>color</i> | <i>clothes_size</i> | <i>quantity</i> |
|------------------|--------------|---------------------|-----------------|
| skirt            | dark         | <b>all</b>          | 8               |
| skirt            | pastel       | <b>all</b>          | 35              |
| skirt            | white        | <b>all</b>          | 10              |
| skirt            | <b>all</b>   | <b>all</b>          | 53              |
| dress            | dark         | <b>all</b>          | 20              |
| dress            | pastel       | <b>all</b>          | 10              |
| dress            | white        | <b>all</b>          | 5               |
| dress            | <b>all</b>   | <b>all</b>          | 35              |
| shirt            | dark         | <b>all</b>          | 14              |
| shirt            | pastel       | <b>all</b>          | 7               |
| shirt            | White        | <b>all</b>          | 28              |
| shirt            | <b>all</b>   | <b>all</b>          | 49              |
| pant             | dark         | <b>all</b>          | 20              |
| pant             | pastel       | <b>all</b>          | 2               |
| pant             | white        | <b>all</b>          | 5               |
| pant             | <b>all</b>   | <b>all</b>          | 27              |
| <b>all</b>       | dark         | <b>all</b>          | 62              |
| <b>all</b>       | pastel       | <b>all</b>          | 54              |
| <b>all</b>       | white        | <b>all</b>          | 48              |
| <b>all</b>       | <b>all</b>   | <b>all</b>          | 164             |



# Extended Aggregation to Support OLAP

- The **cube** operation computes union of **group by**'s on every subset of the specified attributes

- Example relation for this section

*sales(item\_name, color, size, quantity)*

- E.g., consider the query

```
select item_name, color, size, sum(quantity)
from sales
group by cube(item_name, color, size)
```

This computes the union of eight different groupings of the *sales* relation:

{ (*item\_name, color, size*),      (*item\_name, color*),  
  (*item\_name, size*),            (*color, size*),  
  (*item\_name*),                (*color*),  
  (*size*),                      ( ) }

where ( ) denotes an empty **group by** list.

- For each grouping, the result contains the null value for attributes not present in the grouping.





## Extended Aggregation (Cont.)

- The **rollup** construct generates union on every prefix of specified list of attributes
- E.g.,  

```
select item_name, color, size, sum(quantity)
from sales
group by rollup(item_name, color, size)
```

  - Generates union of four groupings:  
 $\{ (item\_name, color, size), (item\_name, color), (item\_name), ( ) \}$
- Rollup can be used to generate aggregates at multiple levels of a hierarchy.
  - E.g., suppose table *itemcategory*(*item\_name*, *category*) gives the category of each item. Then  

```
select category, item_name, sum(quantity)
from sales, itemcategory
where sales.item_name = itemcategory.item_name
group by rollup(category, item_name)
```

would give a hierarchical summary by *category* and by *item\_name*.



## Extended Aggregation (Cont.)

- Multiple rollups and cubes can be used in a single group by clause
  - Each generates set of group by lists, cross product of sets gives overall set of group by lists
- E.g.,

```
select item_name, color, size, sum(number)
from sales
group by rollup(item_name), rollup(color, size)
```

generates the groupings

$$\{item\_name, ()\} \times \{(color, size), (color), ()\}$$
$$= \{ (item\_name, color, size), (item\_name, color), (item\_name), (color, size), (color), () \}$$



# Online Analytical Processing Operations

- **Pivoting:** changing the dimensions used in a cross-tab
- **Slicing:** creating a cross-tab for fixed values only
  - Sometimes called **dicing**, particularly when values for multiple dimensions are fixed.
- **Rollup:** moving from finer-granularity data to a coarser granularity
- **Drill down:** The opposite operation - that of moving from coarser-granularity data to finer-granularity data



# OLAP Implementation

- The earliest OLAP systems used multidimensional arrays in memory to store data cubes, and are referred to as **multidimensional OLAP (MOLAP)** systems.
- OLAP implementations using only relational database features are called **relational OLAP (ROLAP)** systems
- Hybrid systems, which store some summaries in memory and store the base data and other summaries in a relational database, are called **hybrid OLAP (HOLAP)** systems.



## End of Chapter 5